

## Factors



1 State whether the following are factors of  $33wy$ :

a 11

b  $66wy$

c  $yw$

d  $3w$

e  $yw^2$

f 33

2 Find the highest common factor of the following groups of numbers:

a 2 and  $6x$

b 3 and  $15x$

c  $6x$  and 21

d 10 and  $16u$

e  $35x$  and 14

f  $4x$  and  $16xy$

g 15 and  $5x$

h  $24m$ ,  $12n$  and  $18q$

i  $27m$ ,  $24n$  and  $15q$

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## Factorisation

3 Complete the factorisation of the following expressions:

a  $2a + 16 = \square (a + 8)$

b  $5y - 30 = \square (y - 6)$

c  $24n + 15 = \square (8n + 5)$

d  $16a - 6 = \square (8a - 3)$

e  $6v + 18 = 6 (\square)$

f  $9a - 36 = 9 (\square)$

g  $24a + 9 = 3 (\square)$

h  $6w - 27 = 3 (\square)$

i  $4r + 12 = \square (r + 3)$

j  $5v - 30 = \square (v - 6)$

k  $18c - 24 = \square (3c - 4)$

l  $2r + 14 = 2 (\square)$

m  $5t - 10 = 5 (\square)$

n  $54u + 12 = 6 (\square)$

4 Factorise the following expressions:

a  $6v + 30$

b  $6a - 48$

c  $45t - 40$

d  $2m + 10n$

e  $54 - 60fg$

f  $8g - 72h$

g  $9f + 15g$

h  $3w + 15$

i  $2c - 18$

j  $40c - 35$

k  $6w + 30y$

l  $14y + 4w$

m  $4a - 14b$

n  $32 - 36jk$

o  $8jkt - 18$

p  $3fg - 10fh$

q  $3rtw - 8rw$

r  $4rt - 6r^2t$

s  $16wy - 36w^2y$

5 Complete the following factorisation:

$$2w - 10z + 8y = 2 (\square - \square + \square)$$

6 Factorise the following completely:



a  $8g + 28h - 20$

c  $4u - 12v + 20w$

e  $3m - 6n + 12p$

g  $zw - zy - zk$

i  $12m - 32mn - 20m^2n$

k  $25r^2t - 15rt^2 - 45r^2t^2$

b  $4w + 4z + 4y$

d  $10m - 25n - 15p$

f  $10g - 35h + 15$

h  $60r - 25rt - 35r^2t$

j  $16z - 32yz - 24y^2z$

## Factorisation of negatives

7 Factor out the negative coefficient of the following expressions:

a  $-4r - 3s$

b  $-7r - 7s$

c  $-2 - 9r$

d  $-8x - 5y$

8 Factorise the following expressions:

a  $-8b - 16$

b  $-18a + 16$

c  $-80v - 30$

d  $-36 - 6x$

e  $-25u - 35v$

f  $-35m + 49n$

g  $-8pqr - 6pr$

h  $-8w^2 + 3w^2y$

i  $-24j^2 - 27j^2k$

j  $-8m^2 + 9mn$

k  $-10u^2v + 9uv^2$

l  $-21j^2 - 36j^2k$

m  $-2s - 10$

n  $-8v^2 + 56$

o  $-12s + 10$

p  $-28b - 21$

q  $-63r^2 + 28$

r  $-4m - 32n$

s  $-49y - 21w$

9 Factorise the following completely:

a  $-18p - 9q - 45r$

b  $-10r - 25s + 20t$

c  $-20u - 25v + 45w$

d  $-8w - 18y - 10z$